

Challenge (Habanero)

Q1. Simplify $\sqrt{8} + \sqrt{18}$

- (A) $2\sqrt{2} + 3\sqrt{2}$
- (B) $4\sqrt{2} + 6\sqrt{2}$
- (C) $6\sqrt{2}$
- (D) $8\sqrt{2}$
- (E) $10\sqrt{2}$

Q2. If a hiker consumes $\frac{1}{4}\sqrt{50}$ pounds of food per mile, how much food will they consume on a $\sqrt{8}$ mile hike?

- (A) $\sqrt{2}$ pounds
- (B) $2\sqrt{2}$ pounds
- (C) $\frac{1}{2}\sqrt{2}$ pounds
- (D) $\frac{1}{2}$ pound
- (E) 2 pounds

Q3. A wildlife conservationist tracks the population of a species, which grows at a rate of $2\sqrt{3}$ per year. If the initial population is $\sqrt{12}$, what will the population be after $\sqrt{3}$ years?

- (A) $4\sqrt{3}$
- (B) $6\sqrt{3}$
- (C) $8\sqrt{3}$
- (D) $10\sqrt{3}$
- (E) $12\sqrt{3}$

Q4. Simplify $\sqrt{20} - \sqrt{5}$

- (A) $\sqrt{5}$
- (B) $2\sqrt{5}$
- (C) $3\sqrt{5}$
- (D) $4\sqrt{5}$
- (E) $\sqrt{15}$

Q5. A group of friends plan a road trip that is $\sqrt{32}$ miles long. If their car consumes $\frac{1}{2}\sqrt{2}$ gallons of fuel per mile, how many gallons will they consume on the trip?

- (A) 4 gallons
- (B) 8 gallons
- (C) 12 gallons
- (D) 16 gallons
- (E) 20 gallons

Q6. Simplify $\sqrt{48} \cdot \sqrt{12}$

- (A) 24
- (B) 48
- (C) $24\sqrt{2}$
- (D) $48\sqrt{2}$
- (E) 96

Q7. A park ranger measures the length of a trail to be $\sqrt{50}$ miles. If the trail is $\frac{1}{5}\sqrt{10}$ miles wide, what is the area of the trail in square miles?

- (A) $2\sqrt{5}$ square miles
- (B) $5\sqrt{5}$ square miles
- (C) $10\sqrt{5}$ square miles
- (D) $20\sqrt{5}$ square miles
- (E) $50\sqrt{5}$ square miles

Q8. Simplify $\frac{\sqrt{72}}{\sqrt{8}}$

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) $3\sqrt{2}$

Open Ended Questions (FRQ)

Q9. A group of hikers are planning a trip to a mountain that is $\sqrt{98}$ miles away. If they hike at a rate of $2\sqrt{2}$ miles per hour, and they consume $\frac{1}{4}\sqrt{5}$ pounds of food per hour, how many pounds of food will they consume on the trip? Show your work and simplify your answer.

Q10. A wildlife conservationist is tracking the population of a species that grows at a rate of $\sqrt{24}$ per year. If the initial population is $\sqrt{16}$, what will the population be after 3 years? Show your work and simplify your answer.

Chef's Tips

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- Tip 1: Remind students to simplify their radicals whenever possible.
- Tip 2: Encourage students to use estimation to check their answers.
- Tip 3: Emphasize the importance of showing work and explaining reasoning.